

5 Data Exploration

Descriptive statistics

```
## Warning: package 'tidyverse' was built under R version 4.5.2
```

```
#This is the definitive API dataset
efteling_live_queue <- read_csv("../data/final/efteling_live_queue.csv")

#This is the definitive web scraped dataset with feature engineering
efteling_his_complete <- read_csv("../data/final/efteling_his_complete.csv")
```

Historical data

```
#Summarise definitive datasets
summary(efteling_live_queue)
```

```
##      timestamp          attraction_name      status
##  Min.   :2025-11-13 16:04:13 Length:8215      Length:8215
## 1st Qu.:2025-11-16 10:37:58 Class :character Class :character
## Median :2025-11-19 12:37:39 Mode  :character Mode  :character
## Mean   :2025-11-20 01:28:41
## 3rd Qu.:2025-11-24 12:49:49
## Max.   :2025-11-27 17:46:51
##
##      queue_type      wait_time
## Length:8215      Min.   : 0.000
## Class :character 1st Qu.: 0.000
## Mode  :character Median : 5.000
##                      Mean   : 9.251
##                      3rd Qu.:15.000
##                      Max.   :65.000
##                      NA's   :1319
```

The summary of the `efteling_live_queue` dataset shows that it contains 8,215 observations, with waiting times ranging from 0 (min. `wait_time`) to 65 minutes (max. `wait_time`). The median waiting time is 5 minutes, which indicates that at least half of the reported queues were very short. The mean is 9.251, which is slightly higher, because of some outliers of longer waiting time. The dataset includes 1,319 missing values within the `wait_time` variable. This means careful treatment of missing data in subsequent analysis is important.

```
## # A tibble: 12 x 5
##   month      avg_crowd_percent avg_temp_actual avg_rain_actual avg_wind_actual
```

##	<ord>	<dbl>	<dbl>	<dbl>	<dbl>
##	1 January	34.3	4.87	0.126	4.87
##	2 February	28.2	7.53	0.0887	4.67
##	3 March	35.6	10.4	0.101	4.67
##	4 April	50.9	13.1	0.0918	4.61
##	5 May	51.4	17.6	0.133	4.23
##	6 June	57.0	21.4	0.057	4.27
##	7 July	45.8	21.2	0.105	3.89
##	8 August	59.8	21.5	0.0538	3.64
##	9 September	41.3	19.2	0.107	4.11
##	10 October	50.0	14.1	0.103	3.81
##	11 November	40.1	9.20	0.127	4.14
##	12 December	45.0	6.82	0.149	4.82

The monthly aggregate of the historical dataset indicates a seasonal pattern in both the crowd levels and the weather conditions. The data shows that average crowd percentages are lowest during winter months, such as January and February, and this percentage steadily increases towards the summer, with peaks in June and August. These peaks in crowd percentages coincide with higher average temperatures (avg_temp_actual), which indicates that warmer weather is associated with a higher number of park visitors. Rainfall levels (avg_rain_actual) does not show a strong seasonal pattern, although somewhat greater values are observed in the spring and fall. Wind conditions (avg_wind_actual) remain relatively stable during the year. Overall, these results indicate that seasonal weather conditions are correlated with fluctuations in park crowding.

```
## # A tibble: 12 x 3
##   month      avg_crowd_percent avg_queue_time
##   <ord>          <dbl>          <dbl>
## 1 January         40.4            14.3
## 2 February        30.6            11.6
## 3 March           37.8            13.2
## 4 April           51.4            15.9
## 5 May             51.9            16.4
## 6 June            56.6            17.2
## 7 July            45.9            14.8
## 8 August          60.3            18.0
## 9 September       42.5            14.5
## 10 October        50.9            15.6
## 11 November       43.1            15.8
## 12 December       48.8            16.8
```

Where in the previous table the focus was laid upon the monthly crowd levels in relation to weather conditions, this summary examines the relation between crowd levels and queue times.

The monthly averages reveal a seasonal pattern in crowd levels and queue times. Avg_crowd_percentages rise during the spring and summer months, with peaks in June and August. The increase in park attendance coincide with longer avg_queue_times. In contrast, lower crowd levels are noticeable in winter months, such as January and February, which correspond with shorter waiting times. Overall, these results indicate a strong positive correlation between the monthly crowd percentages and avg_queue_times, which indicates that periods that are busier consistently lead to longer queues in the park.

Live data

```
#Summarise definitive datasets
summary(efteling_his_complete)
```

```
##      date      park_id      ride      avg_queue_min
## Min.   :2023-01-01  Min.   :160  Length:31342  Min.   : 5.00
## 1st Qu.:2023-09-28  1st Qu.:160  Class :character 1st Qu.: 7.00
## Median :2024-06-24  Median:160  Mode  :character Median :13.00
## Mean   :2024-06-17  Mean   :160                Mean   :15.44
## 3rd Qu.:2025-03-09  3rd Qu.:160                3rd Qu.:22.00
## Max.   :2025-11-16  Max.   :160                Max.   :141.00
##                                     NA's   :5290
## max_queue_min  uptime_pct  crowd_percent  crowd_label
## Min.   : 5.00  Min.   : 0.00  Min.   : 0.00  Length:31342
## 1st Qu.:15.00  1st Qu.:89.00  1st Qu.:23.00  Class :character
## Median :25.00  Median :99.00  Median :43.00  Mode  :character
## Mean   :26.36  Mean   :82.91  Mean   :45.13
## 3rd Qu.:35.00  3rd Qu.:100.00  3rd Qu.:66.00
## Max.   :260.00  Max.   :100.00  Max.   :100.00
## NA's   :5290
## temperature_forecast temperature_actual rain_forecast  rain_actual
## Min.   :-2.80      Min.   :-1.70      Min.   :0.0000  Min.   :0.0000
## 1st Qu.: 8.80      1st Qu.: 9.20      1st Qu.:0.0000  1st Qu.:0.0000
## Median :14.30      Median :14.10      Median :0.0000  Median :0.0000
## Mean   :14.28      Mean   :14.22      Mean   :0.1023  Mean   :0.1011
## 3rd Qu.:20.00      3rd Qu.:19.50      3rd Qu.:0.1000  3rd Qu.:0.0800
## Max.   :34.10      Max.   :34.80      Max.   :2.0900  Max.   :2.2300
## NA's   :2069      NA's   :2069
## wind_forecast  wind_actual  events  weekday_weekend
## Min.   : 0.500  Min.   : 0.900  Length:31342  Length:31342
## 1st Qu.: 2.500  1st Qu.: 3.000  Class :character  Class :character
## Median : 3.500  Median : 4.100  Mode  :character  Mode  :character
## Mean   : 3.672  Mean   : 4.288
## 3rd Qu.: 4.600  3rd Qu.: 5.300
## Max.   :11.200  Max.   :12.400
## NA's   :2069
## has_single_rider indoor_outdoor  capacity_theoretical  opening_year
## Min.   :0.0000  Length:31342  Min.   : 50  Min.   :1956
## 1st Qu.:0.0000  Class :character  1st Qu.:1000  1st Qu.:1982
## Median :0.0000  Mode  :character  Median :1400  Median :1993
## Mean   :0.2134  Mean   :1292  Mean   :1996
## 3rd Qu.:0.0000  3rd Qu.:1800  3rd Qu.:2015
## Max.   :1.0000  Max.   :2010  Max.   :2024
##
## is_school_holiday_any is_holiday
## Mode :logical      Mode :logical
## FALSE:20841      FALSE:30421
## TRUE :10501       TRUE :921
##
##
##
##
```

This definitive data set of the historical data contains 31,342 observations covering the period from January 2023 to November 2025. Average queue times range from 5 (min) to 141 (max) minutes, with 5,290 missing values. Crowd levels vary between the 0 and the 100%, which shows variability throughout the data. Weather variables, like temperature, rainfall, and wind speed vary widely across the data set. Additional contextual information relevant for modelling queue dynamics, such as ride type, or indoor vs. outdoor, are also incorporate in this data set.

```
#Of live data, calculate average queue time
efteling_live_queue %>%
  group_by(attraction_name, queue_type) %>%
  summarise(averagequeue = mean(wait_time, na.rm = TRUE)) %>%
  arrange(desc(averagequeue)) %>%
  print(n = 31)
```

'summarise()' has grouped output by 'attraction_name'. You can override using
the '.groups' argument.

```
## # A tibble: 31 x 3
## # Groups:   attraction_name [31]
##   attraction_name      queue_type averagequeue
##   <chr>              <chr>      <dbl>
## 1 Danse Macabre      STANDBY      25.7
## 2 Stoomtrein Ruigrijk STANDBY      23.8
## 3 Stoomtrein Marerijk STANDBY      23.7
## 4 Symbolica          STANDBY      23.1
## 5 Baron 1898         STANDBY      17.4
## 6 Baron 1898 Single-rider SINGLE_RIDER 15.6
## 7 Joris en de Draak  STANDBY      15.0
## 8 Droomvlucht        STANDBY      14.7
## 9 Python             STANDBY      12.1
## 10 De Vliegende Hollander STANDBY      11.9
## 11 Fabula             STANDBY       9.98
## 12 De Oude Tufferbaan STANDBY       9.42
## 13 Max & Moritz       STANDBY       9.18
## 14 Vogel Rok          STANDBY       7.68
## 15 Fata Morgana       STANDBY       7.09
## 16 Carnaval Festival STANDBY       6.31
## 17 Villa Volta        STANDBY       5.55
## 18 Halve Maen         STANDBY       5.12
## 19 Pagode             STANDBY       4.94
## 20 Joris en de Draak Single-rider SINGLE_RIDER 4.32
## 21 Kinderspoor        STANDBY       2.95
## 22 Monorail           STANDBY       2.94
## 23 Max & Moritz Single-rider SINGLE_RIDER 2.87
## 24 Symbolica Single-rider SINGLE_RIDER 2.84
## 25 Piraña             STANDBY       2.53
## 26 Python Single-rider SINGLE_RIDER 2.16
## 27 Gondoleтта        STANDBY       1.92
## 28 De Vliegende Hollander Single-rider SINGLE_RIDER 0.822
## 29 Stoomcarroussel    STANDBY       0.797
## 30 Danse Macabre Single-rider SINGLE_RIDER 0
## 31 Sirocco           STANDBY      NaN
```

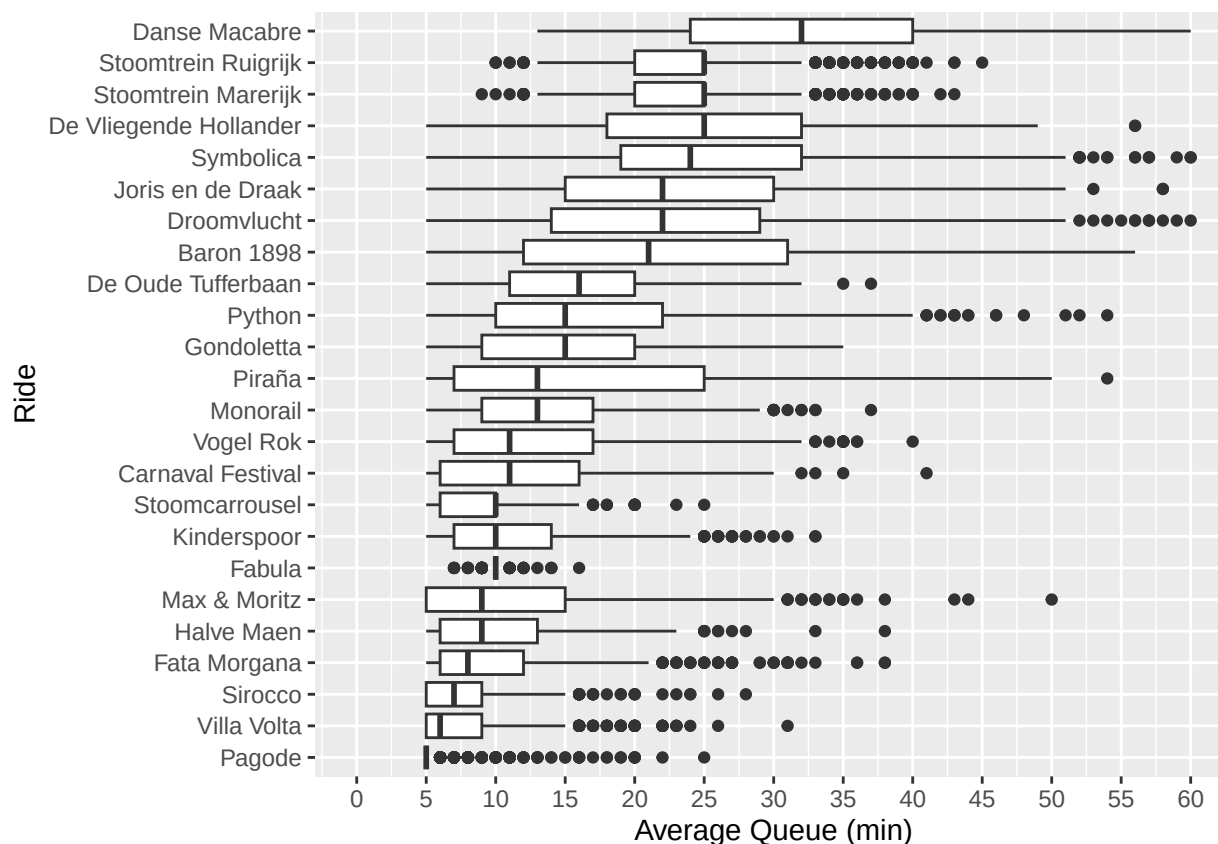
Utilizing the live queue data set, which contains 8,215 observations, the analysis identified all 31 attractions

of which the average waiting time can be calculated. The table shows all the attractions in combination with their corresponding queue type, such as standby or single_rider. This output can be used to compare queue dynamics across rides and queue systems.

Data Visualisations

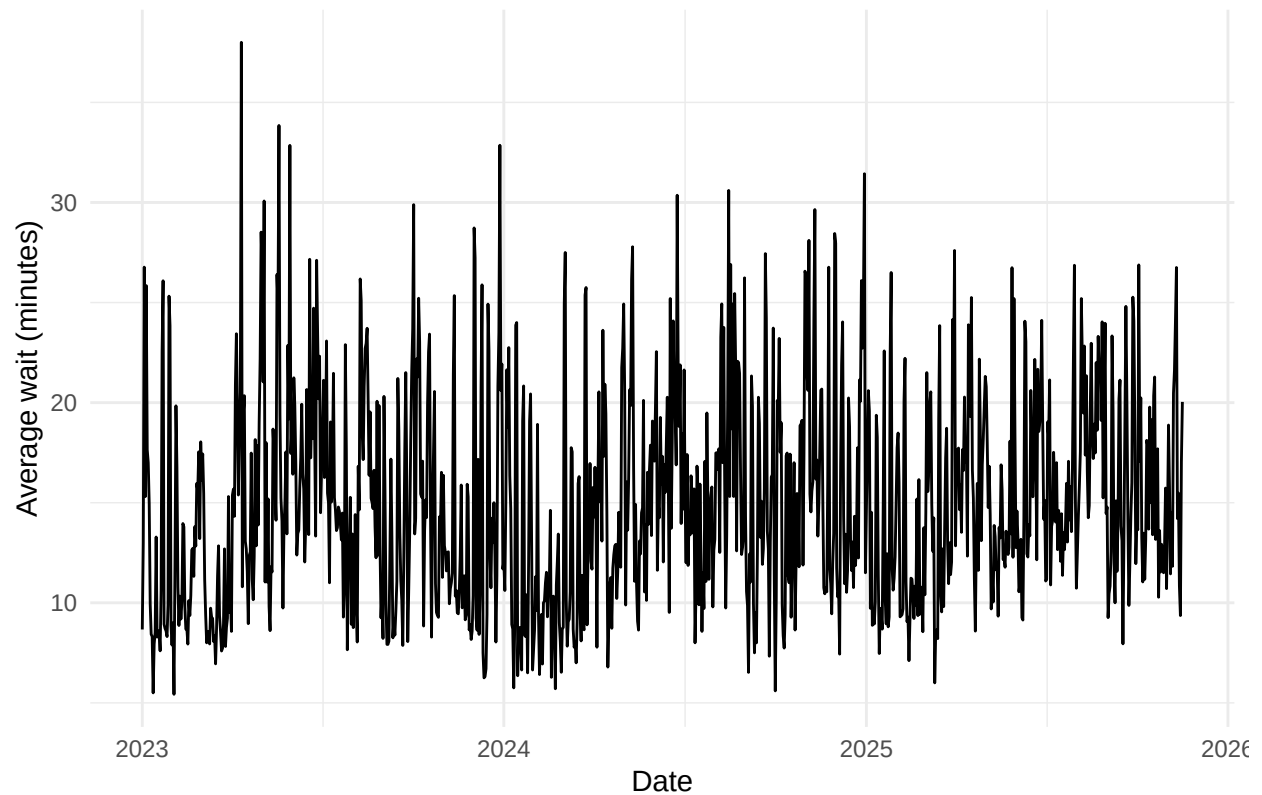
Historical data

```
## Warning: Removed 40 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



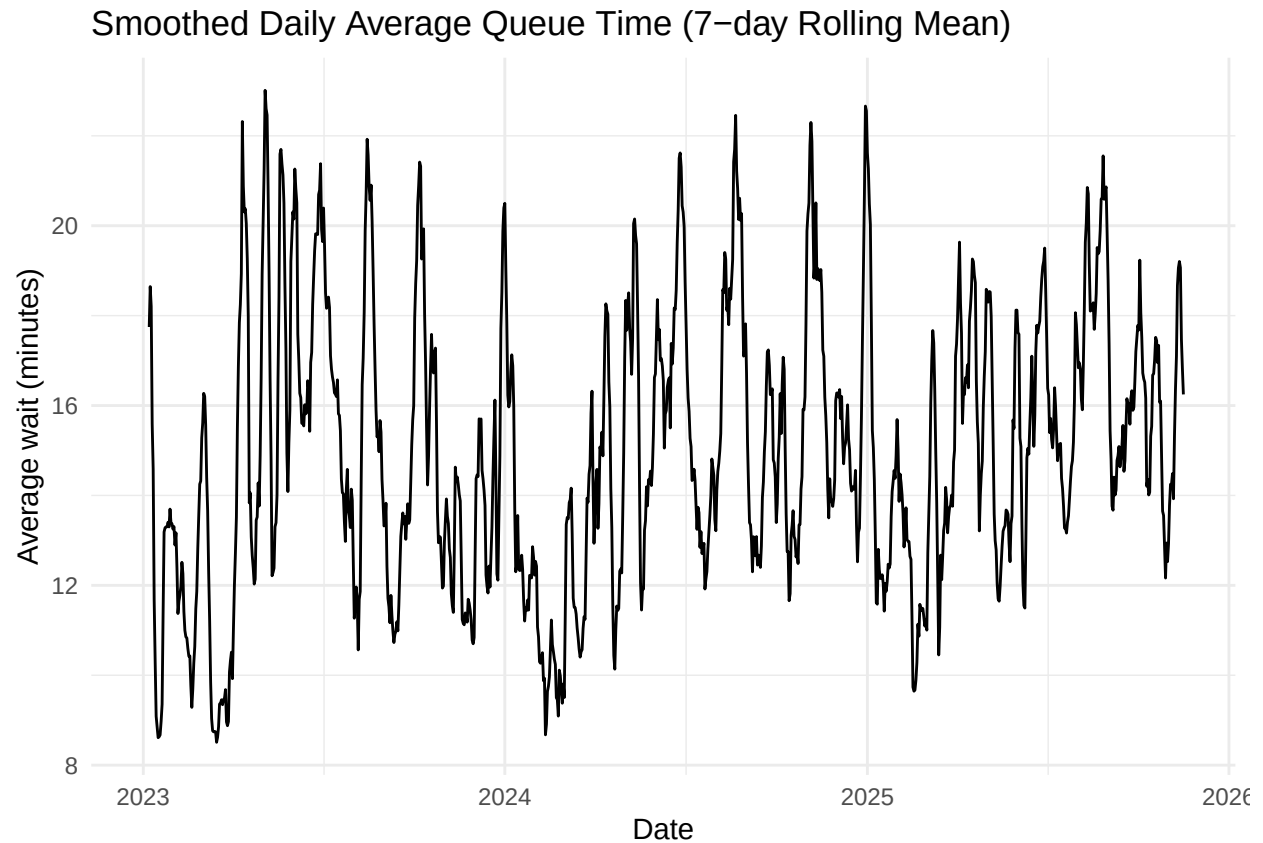
This boxplot shows a summary of the historical queue times for the 31 attractions, with an exclusion on the single-rider lines. A big part of the rides show median waiting times between the 10 and 30 minutes. Several rides, such as Symbolica, Joris en de Draak, and Baron 1898, show variability and occasional peaks above the 40 minutes. The x-axis was limited to 60 minutes for readability, and two rides has a small number of extreme outliers, that ranged above the 60 minutes. Overall, the boxplot shows the differences in queue times across the rides, where extreme rides generally show longer and more fluctuating wait times.

Daily Average Queue Time Over Time

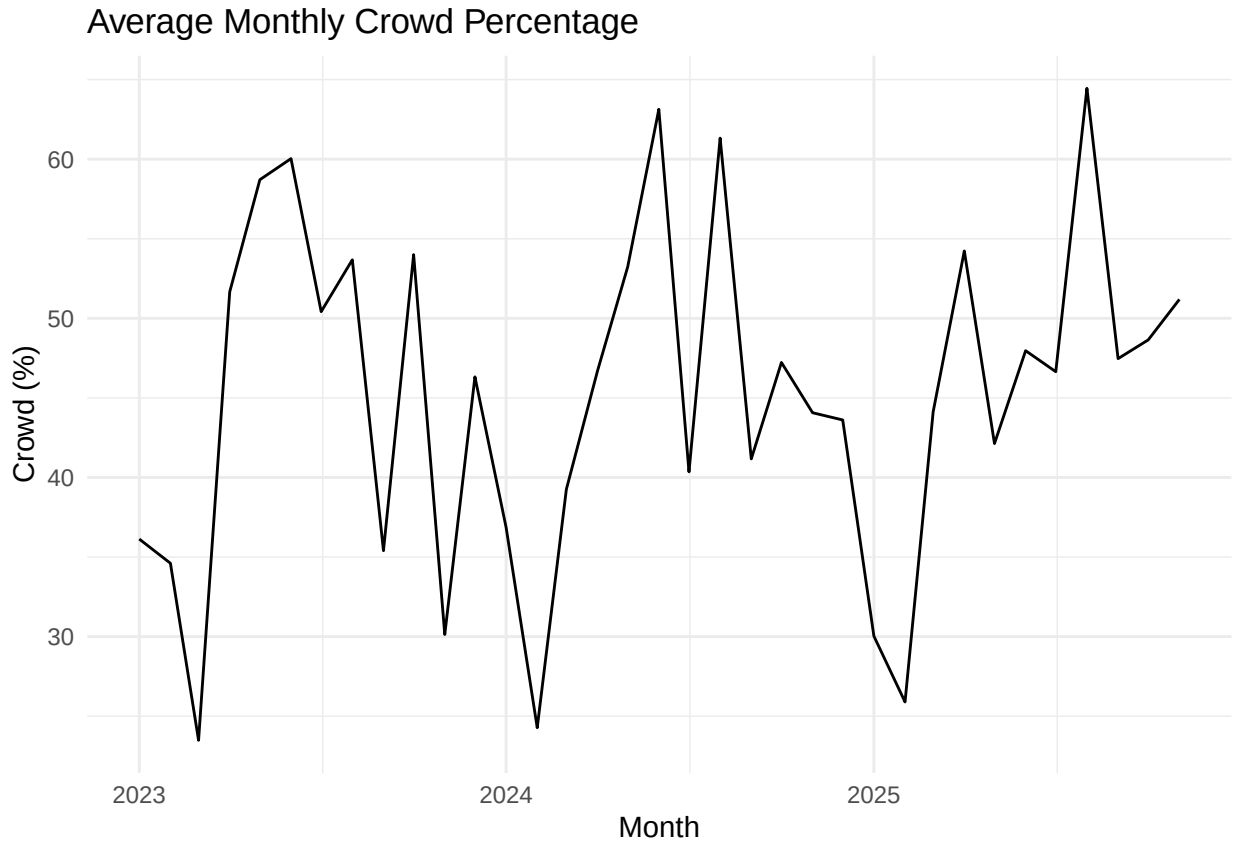


This time-series plot shows the daily average queue time across all attractions. The data fluctuates strongly from day to day, reflecting natural variation due to weather, crowds, holidays, and operational factors. Several clear peaks occur during busy periods, while dips reflect lower-attendance days. Because the raw data is unsmoothed, the figure emphasises how volatile daily queue patterns can be.

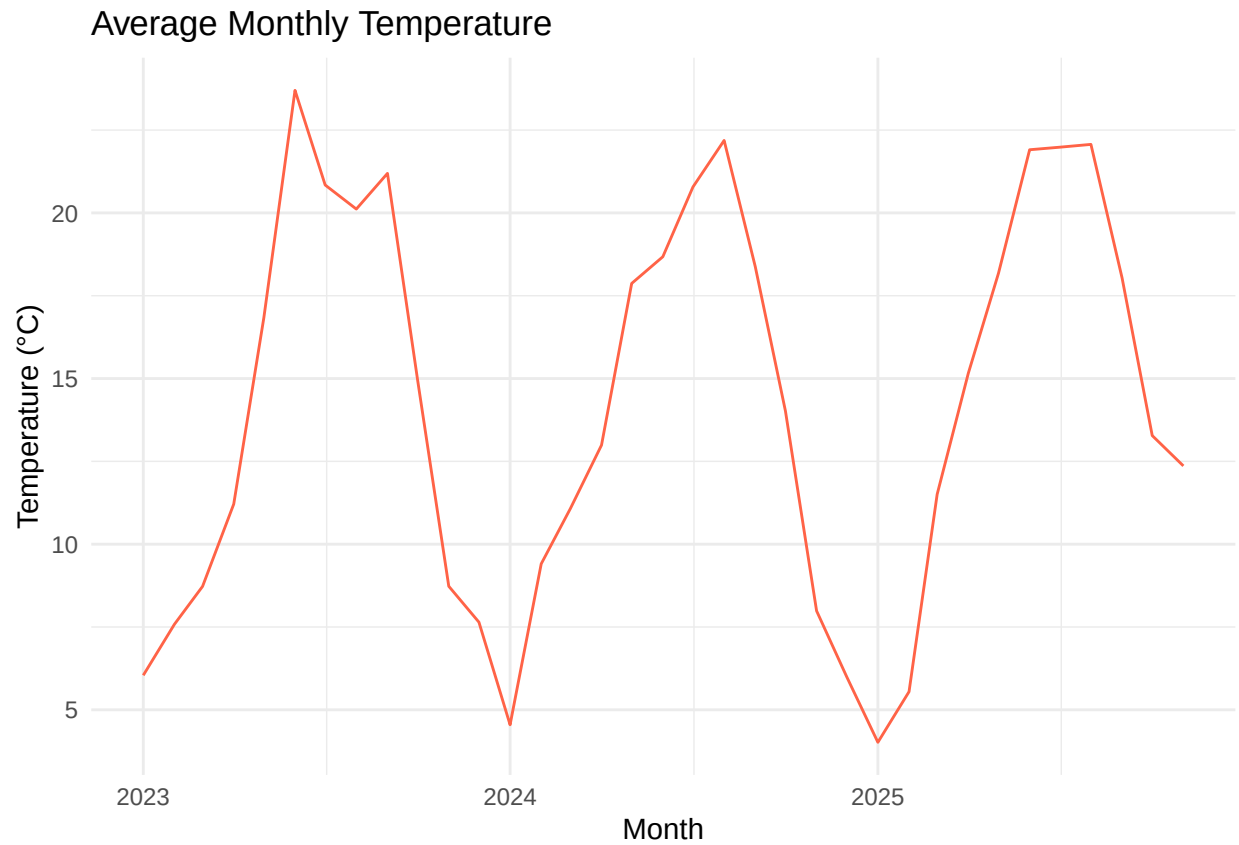
```
## Warning: Removed 6 rows containing missing values or values outside the scale range
## ('geom_line()').
```



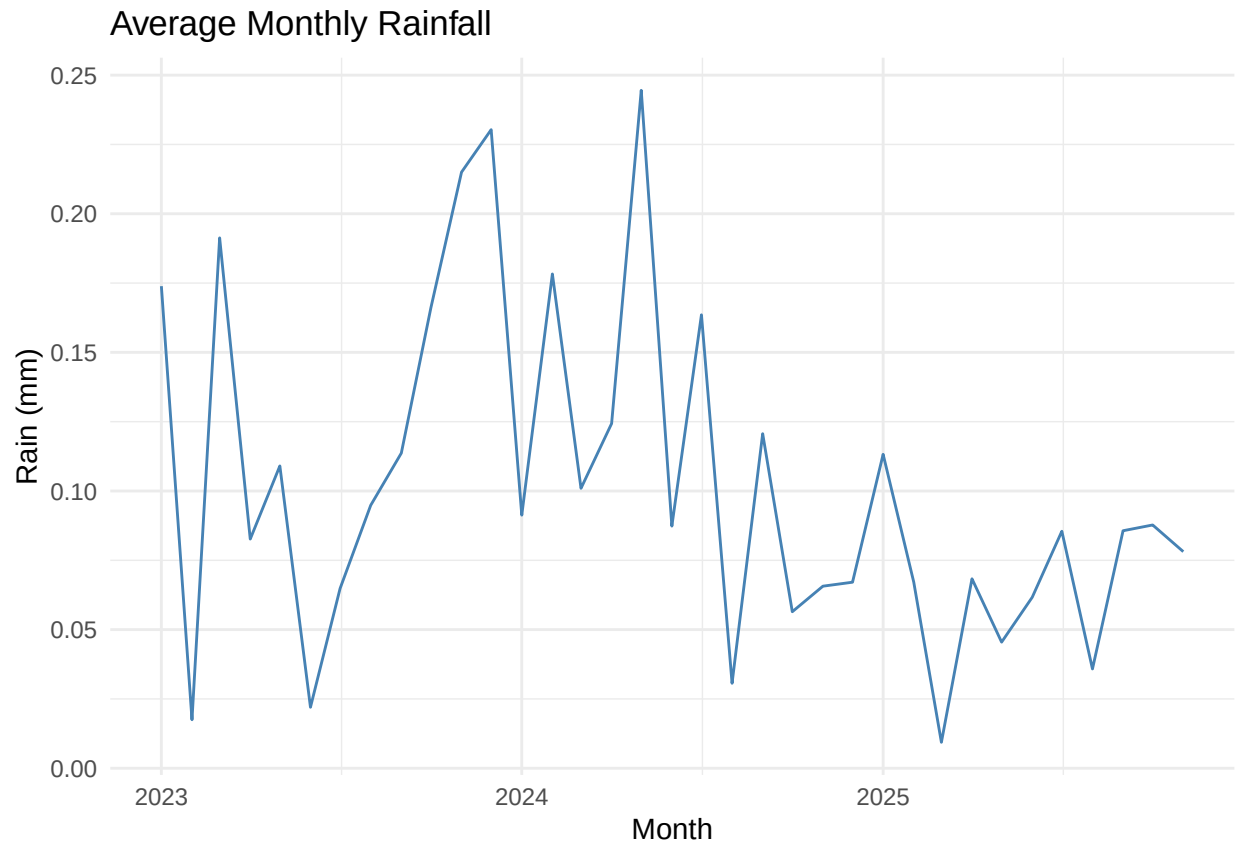
This figure shows the 7-day rolling average of daily queue times, providing a smoother representation of the trend. Short-term fluctuations are reduced, making recurring seasonal patterns easier to observe. Peaks align with known high-attendance periods, such as summer holidays or special events. Compared to the raw daily series, this plot gives a clearer overview of underlying trends in visitor behaviour.



This plot shows the average monthly crowd level across the historical dataset. Crowd percentages rise sharply during spring and summer, particularly around school holidays, and decline during the winter months. These patterns reflect typical theme-park attendance cycles. The monthly view highlights how predictable seasonality influences the general occupancy of the park.

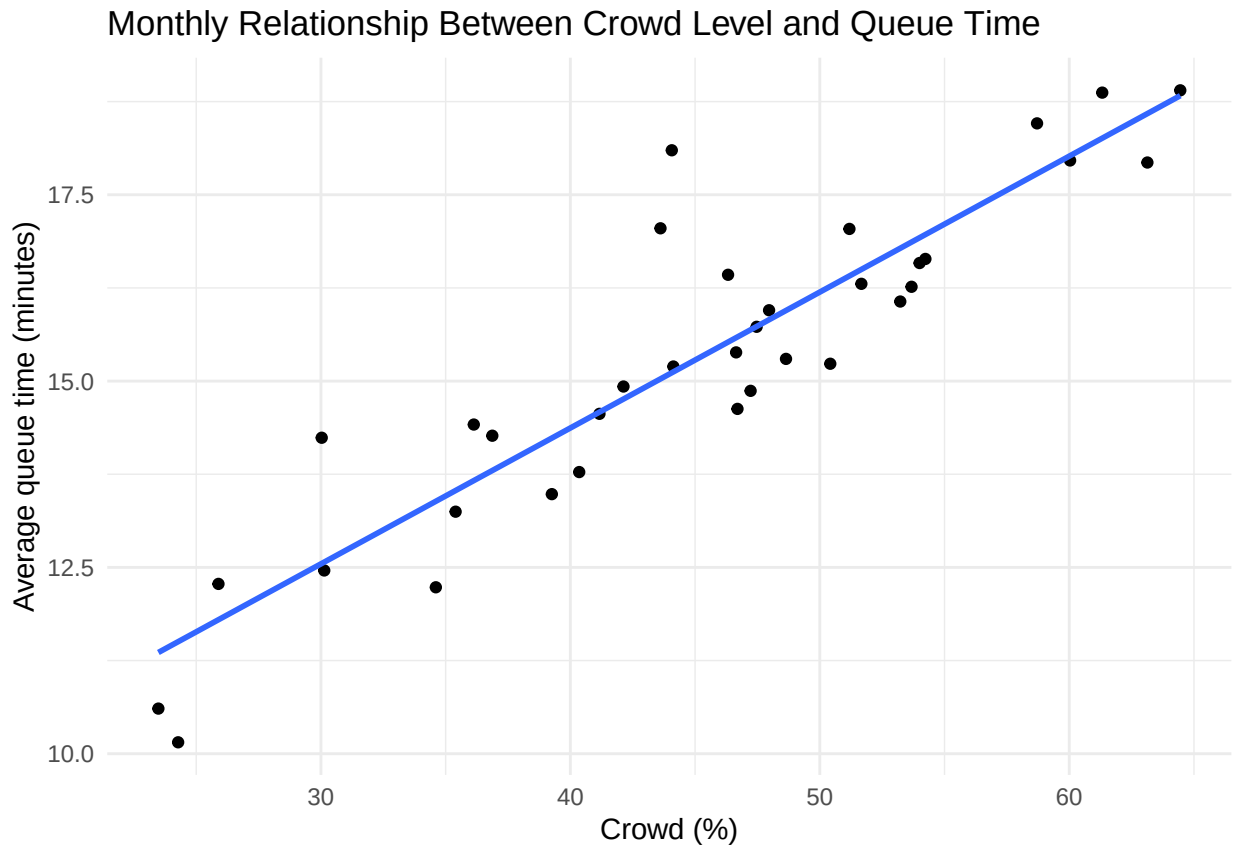


This figure presents the average monthly temperature throughout the year. Temperatures peak during June–August, while the coldest months appear in winter. The seasonal pattern helps contextualise visitor behaviour, as weather conditions can influence both attendance and queue times. Overall, this variable aligns with expected climate trends.

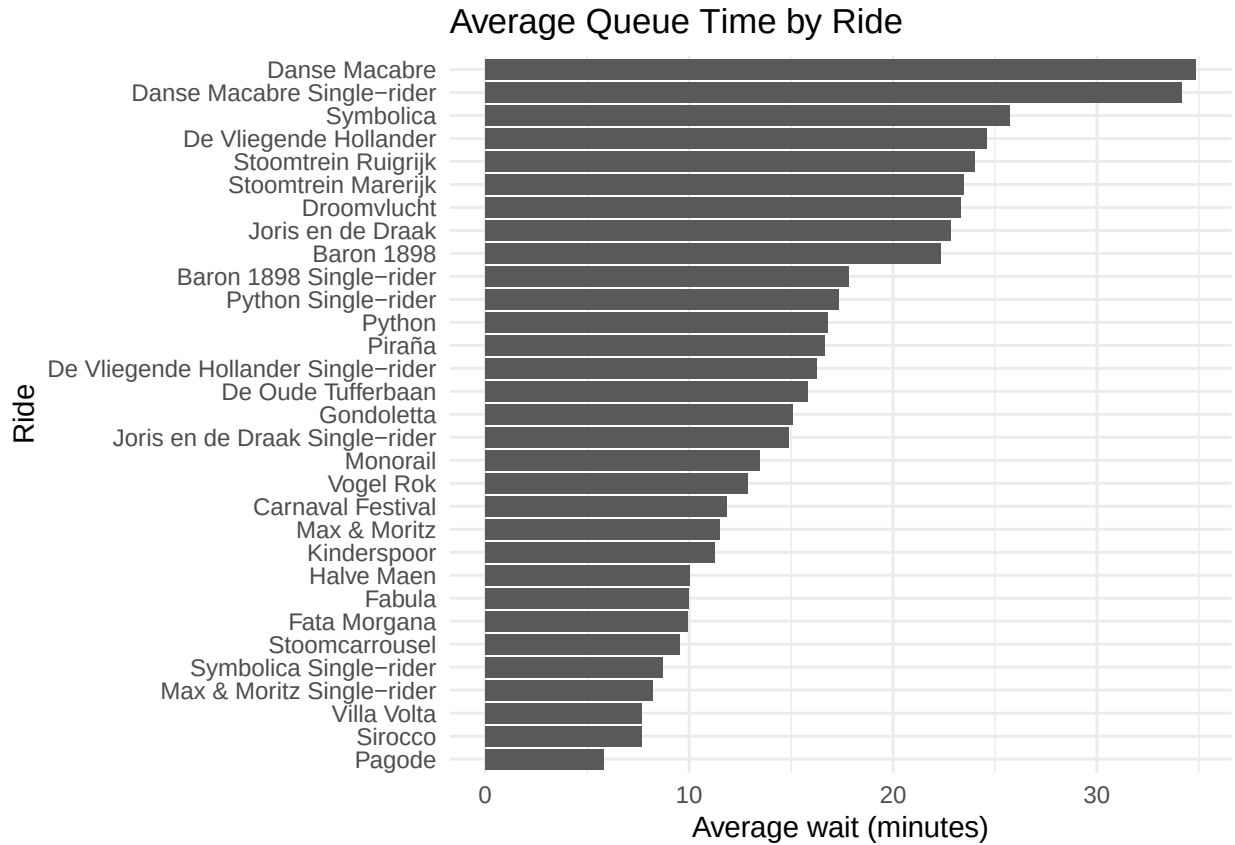


This figure shows the average rainfall per month. Rainfall appears to vary substantially across the year, with wetter months typically occurring in autumn and early spring. Higher rainfall months may correspond to lower attendance levels, although the exact effect varies across days and ride types. The plot helps illustrate how weather conditions fluctuate across seasons.

```
## 'geom_smooth()' using formula = 'y ~ x'
```

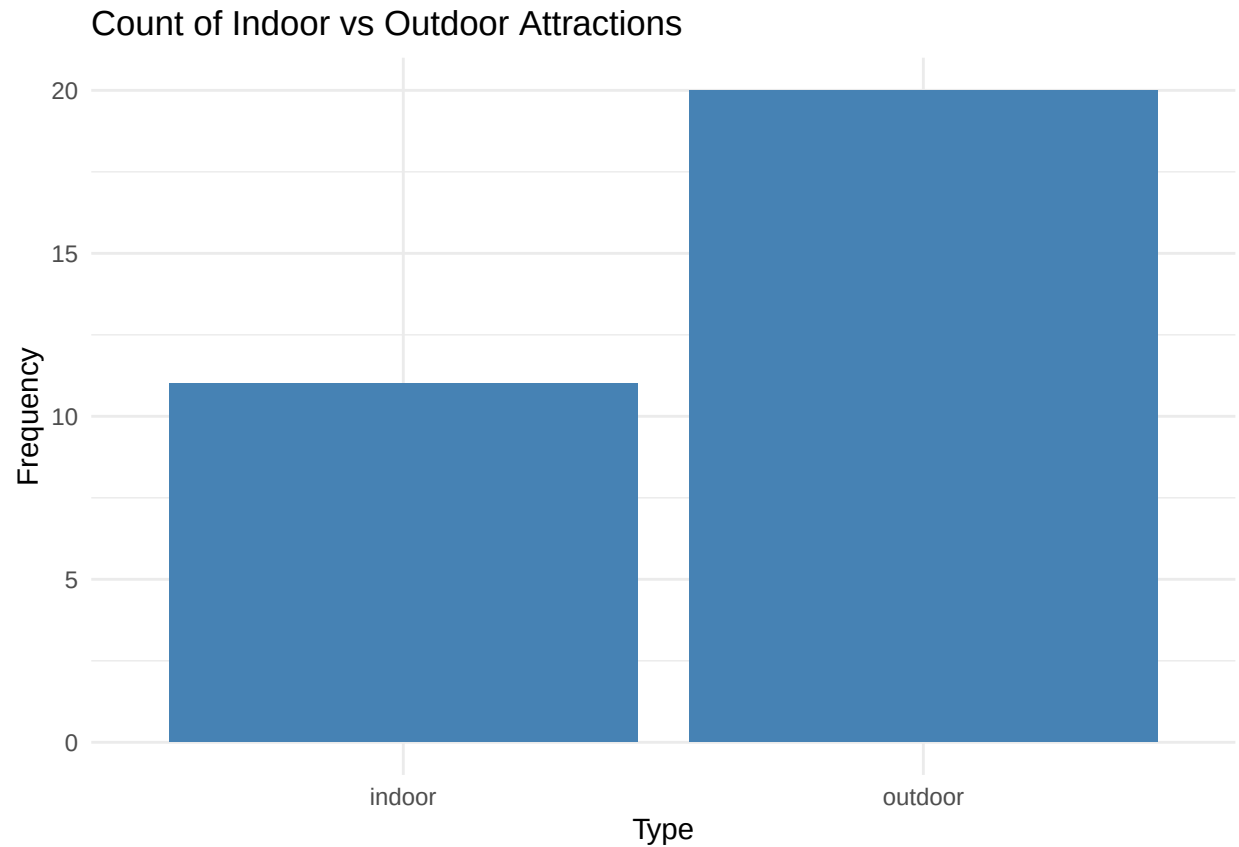


This scatterplot shows the relationship between monthly crowd percentages and average monthly queue times. The clear upward trend indicates that higher crowd levels correspond to longer waiting times. Months with low crowd percentages generally exhibit shorter queues, while highly attended months show substantially longer averages. The fitted regression line emphasises the strength of this positive relationship.



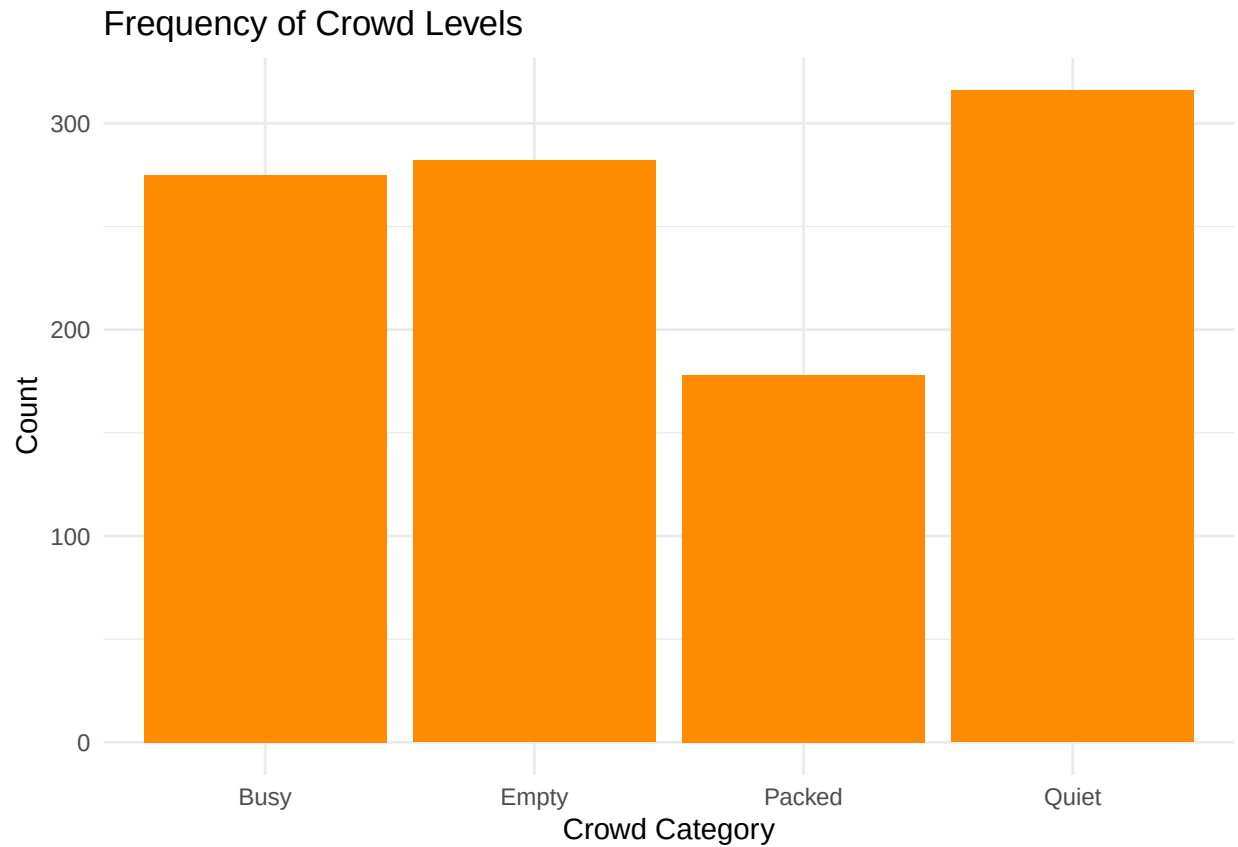
This bar chart shows the average queue time for each attraction across the historical dataset. Some rides, such as Baron 1898, Symbolica, and Droomvlucht, consistently have higher waiting times, indicating strong popularity combined with limited capacity. Other rides show much shorter averages, reflecting either higher throughput or lower demand. The figure provides a clear comparison of typical waiting times across all attractions.

```
##
## indoor outdoor
## 10322 21020
```



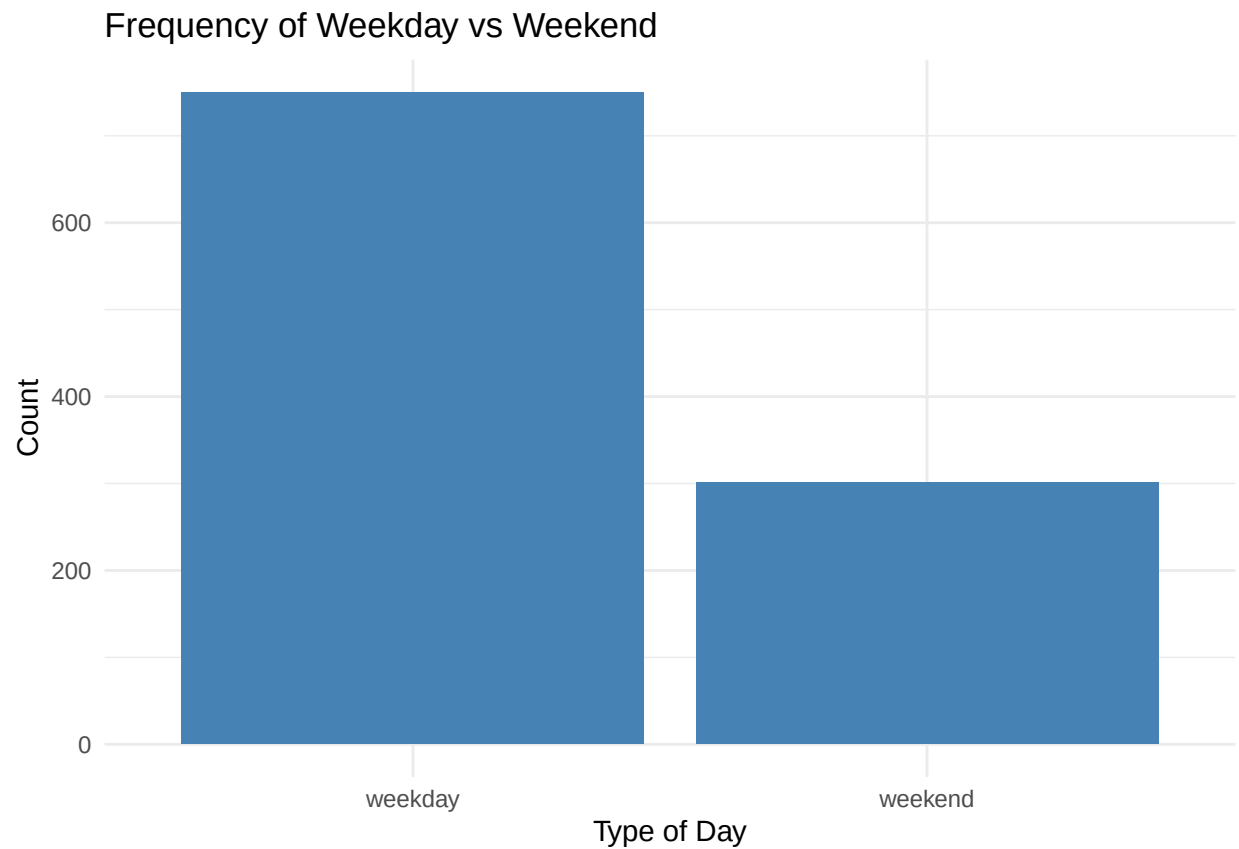
The dataset appears to have nearly twice as many outdoor attractions, around the 20, as indoor attractions, around the 11. This reflects the park's layout, where outdoor rides cover a big part of the overall parkmap. As a consequence, weather conditions are likely to influence the operational performance and waiting times of a big share of the rides.

```
##
##  Busy  Empty Packed  Quiet
##  8213   8375   5293   9461
```



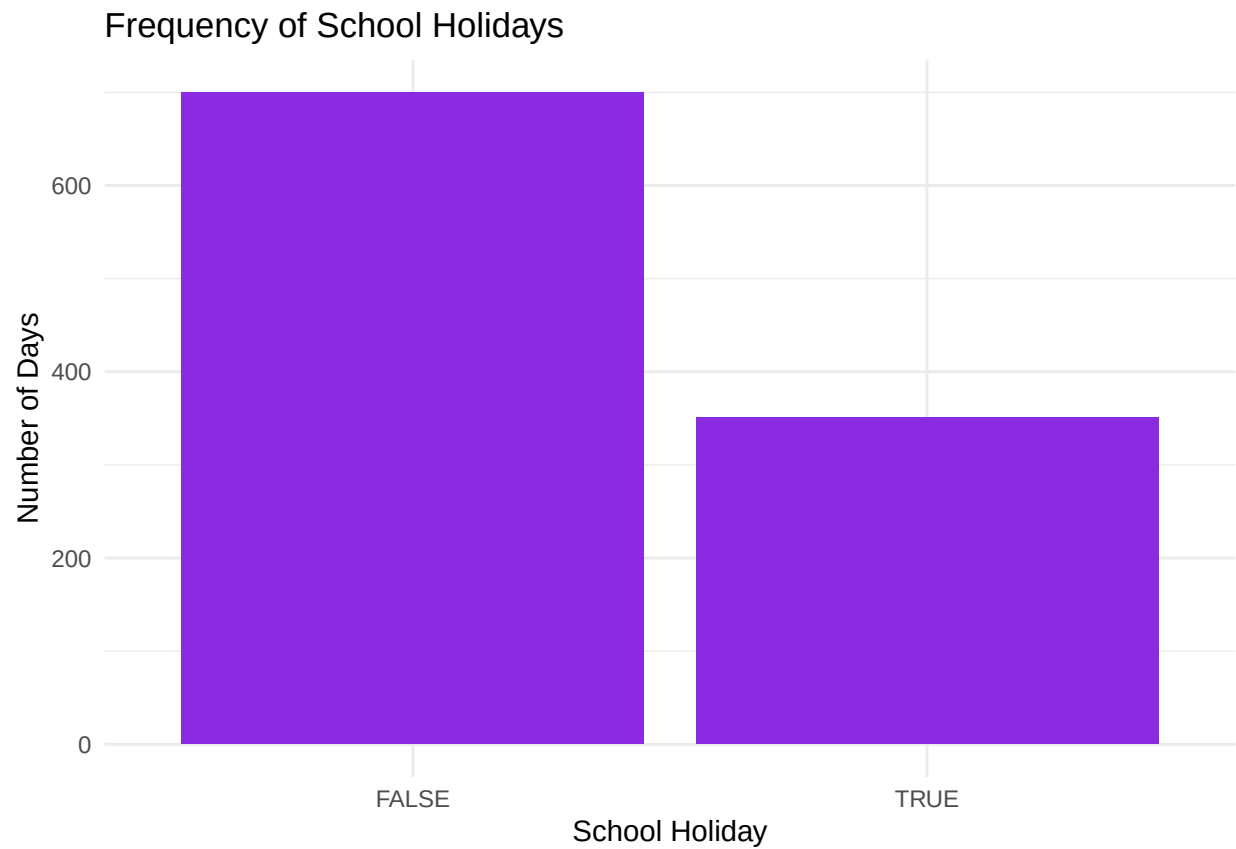
A distribution of the crowd labels show that the “Busy”, “Empty”, and “Quiet” crowd levels occur frequently, with between the 275 and the 320 observations per label. The “packed” conditions occur slightly less frequently. This shows overall, that extreme crowding conditions, such as packed and busy, is relatively less common, than moderate crowding, such as empty and quiet.

```
##  
## weekday weekend  
## 22367 8975
```



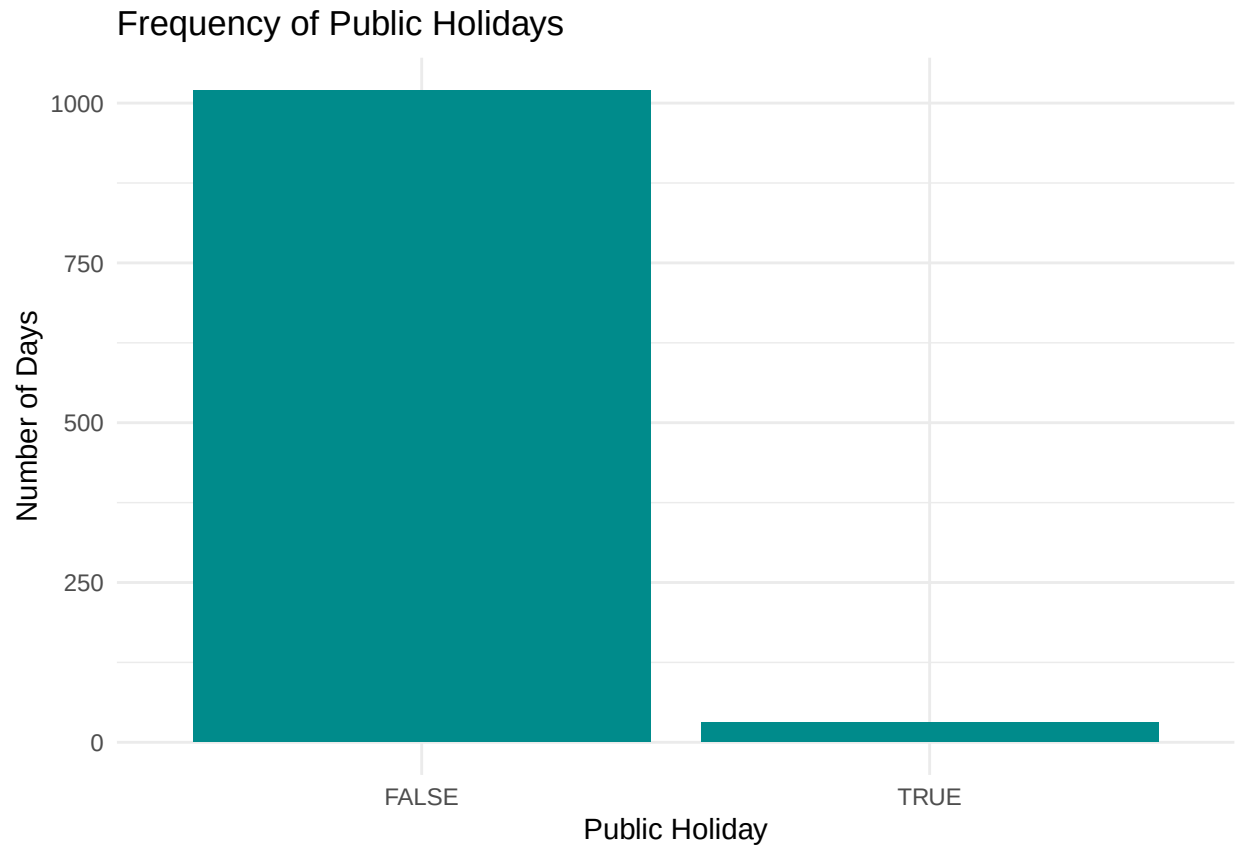
Most observations come from weekdays. This distribution is quite similar to what is expected with the amount of week- and weekenddays in a week.

```
##  
## FALSE TRUE  
## 20841 10501
```



Most observations come from days where there was no school holiday. School holidays are overrepresented in this dataset, as in reality they do not account for one-third of all days.

```
##  
## FALSE  TRUE  
## 30421   921
```

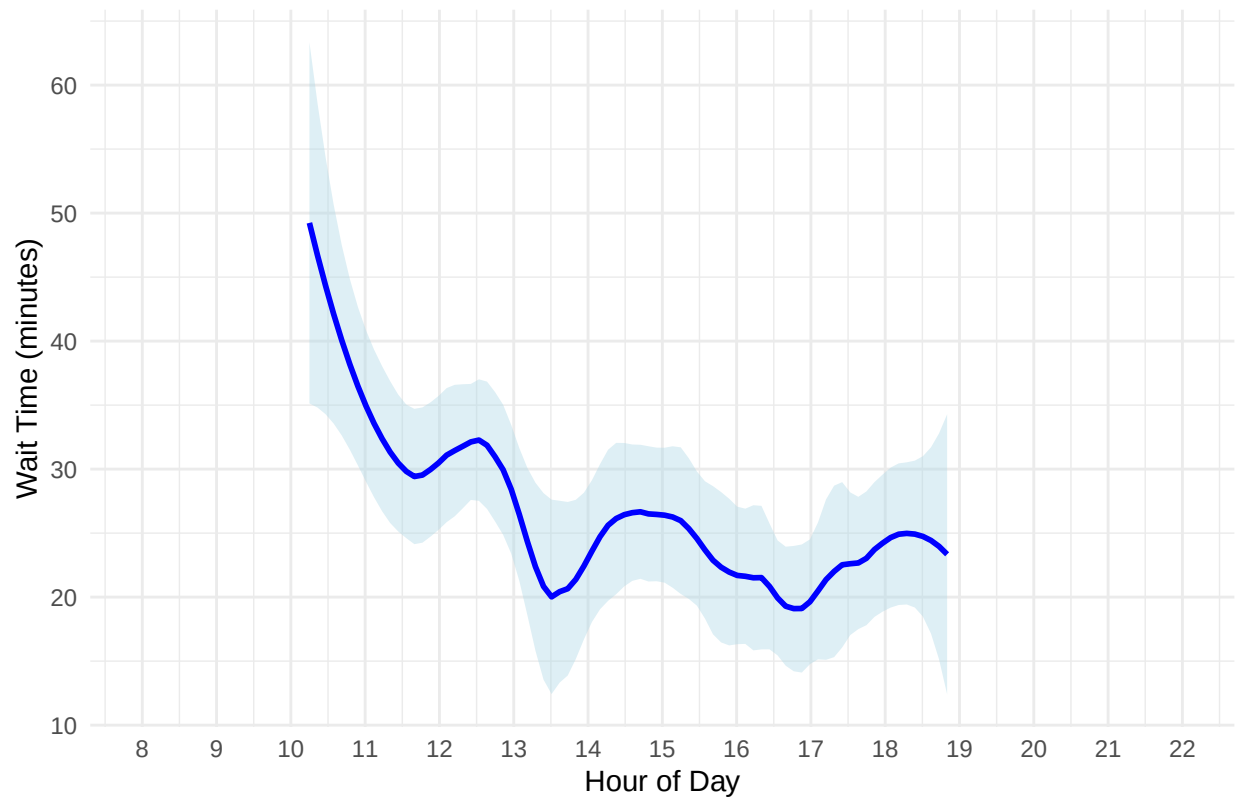
The number of days with a public holiday are only a small part of the total amount of observations. This distribution is similar to real life, since there are only a few public holidays in a year.

Live data

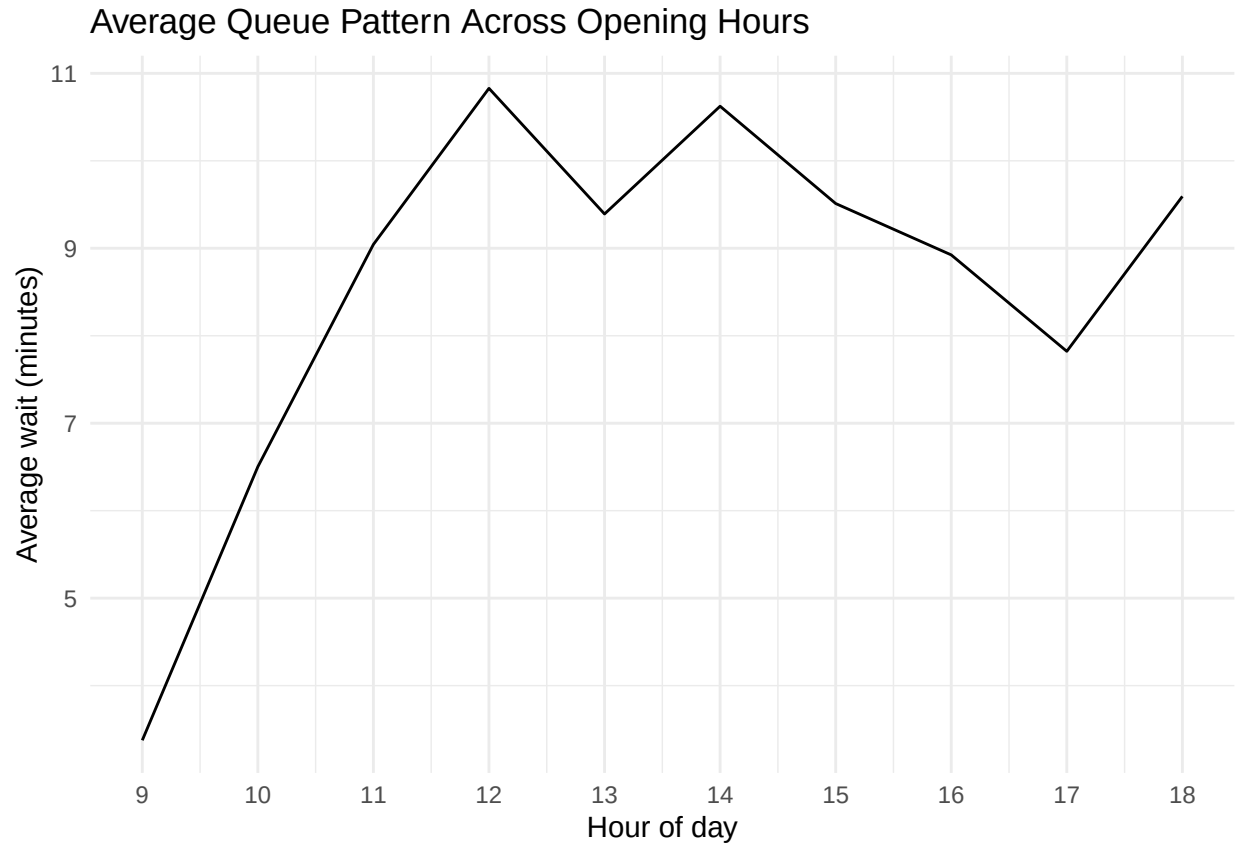
```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 88 rows containing non-finite outside the scale range  
## ('stat_smooth()').
```

Queue Pattern for Danse Macabre Throughout the Day

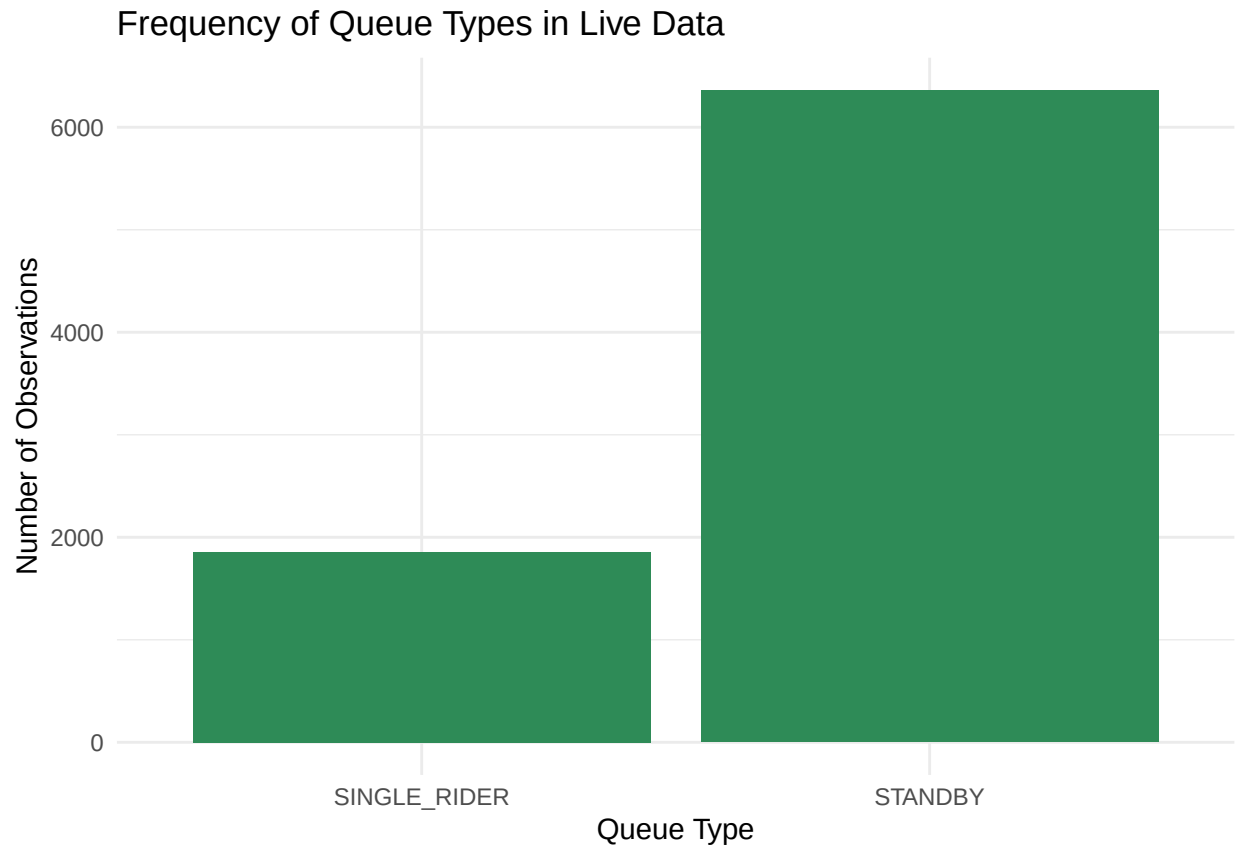


The Queue pattern for Danse Macabre shows a clear decline in waiting times over the day. The queue begins at its highest point around 10 am, where it is approximately 50 - 55 minutes. After this it drops towards the early afternoon. Between 1 pm and 7 pm, the wait time stabilizes at a lower level of 18 - 25 minutes. This daily pattern indicates that the morning period consistently experiences the highest crowd pressure for this attraction.



This figure shows the average live waiting time per hour across all attractions. Queue lengths increase rapidly, shortly after opening and reach their highest levels around midday and early afternoon. Towards the evening, the average waiting time gradually declines as crowds disperse or leave the park. The plot reflects the typical daily rhythm of visitor behaviour in a theme-park environment.

```
##
## SINGLE_RIDER      STANDBY
##          1855      6360
```



Most of the observations within the dataset correspond to the standby queues, with over 6,000 recorded entries. The single-rider queues are less represented with less than 2,000 recorded entries. The live queue data primarily captured the standard queue behavior.

```
## # A tibble: 10 x 2
##   attraction_name      mean_wait
##   <chr>              <dbl>
## 1 Danse Macabre       25.7
## 2 Stoomtrein Ruigrijk  23.8
## 3 Stoomtrein Marerijk  23.7
## 4 Symbolica           23.1
## 5 Baron 1898          17.4
## 6 Baron 1898 Single-rider 15.6
## 7 Joris en de Draak    15.0
## 8 Droomvlucht          14.7
## 9 Python              12.1
## 10 De Vliegende Hollander 11.9
```

This overview indicated that Danse Macabre, the steam trains, and Symbolica have the highest average waiting times, where the means all lay above the 23 minutes. Baron 1898, Joris en de Draak, and Python, the classic thrill rides, experience lower mean waiting times, generally between the 12 and 17 minutes. The overall distribution indicates a variation in attraction popularity.